

COSUCIN POWDER

Description and Composition

COSUCIN POWDER is a light yellow to yellow powder. Each gram contains 125mg Trimethoprim and 625mg Sulfachloropyridazine sodium as its active ingredients.

Pharmacodynamics

The antibacterial activity of veterinary medicinal products is based on a double sequential inhibition of the microbial synthesis of tetrahydrofolic acid. Trimethoprim is an antibacterial derivative of diaminopyrimidine with bactericidal and synergistic activity with sulfonamides. Sulfachloropyridazine is an antibacterial sulfonamide that works by preventing bacterial growth. The combination of the two active substances has a synergistic effect, which influences the dosage (reduction in the doses of the active substances) and the bactericidal effect in a positive way and widens the spectrum of activity. Furthermore, the onset of bacterial resistance is slowed down. In chickens, the veterinary medicinal product is active against respiratory and septic infections by *E. coli* wild type and against septic infections by *P. multocida* and *S. aureus*.

Resistance develops slowly by means of transferable elements and results from a chromosomal mutation following one or more of the mechanisms described below:

- permeability barrier and / or active efflux;
- insensitive natural target enzymes;
- adjustable modifications of the target enzyme;
- mutational or recombinant modifications of the target enzyme;
- resistance acquired by target enzymes resistant to the veterinary medicinal product.

Pharmacokinetics

The pharmacokinetic parameters in chickens after a single administration of the veterinary medicinal product at the recommended daily dose are summarized in the table below, as well as the “steady state” plasma concentrations (C_{ss}) by continuous medication in drinking water. Significantly higher plasma concentrations are achieved with sulfachloropyridazine.

	Trimethoprim	Sulfachloropyridazine
VS _{max} (µg / ml)	0.52	18.65
AUC _{24h} (µg.h / ml)	2.14	103.4
T _{max} (h)	1.55	1.32
T _{1 / 2el} (h)	1.41	2.60
VS _{ss} (µg / ml)	0.089	4.31

Protein fixation in plasma reaches 24.1% for trimethoprim and 35.1% for sulfachloropyridazine. These results translate into an effective C_{ss} (steady state) of 0.068µg/ml for trimethoprim and 2.797µg/ml for sulfachloropyridazine. The two active components are about absorbed as quickly and excreted relatively quickly.

Indication

Treatment of infections caused by bacteria sensitive to trimethoprim / sulfachloropyridazine, taking into account the ability of the antibiotic to reach the site of infection in effective concentrations.

Recommended Dose

To be administered orally through drinking water.

Dosage:

32mg of product per kg of body weight per day (equivalent to 24mg of active substance in total). Particularly in the case of primary and secondary infections by *E. coli*, a higher dose is recommended.

Administration mode:

Chickens: Mix the product in drinking water for 3 to 6 days.

$$\frac{\text{....mg product / kg}}{\text{body weight / day *}} \times \frac{\text{average body weight (kg)}}{\text{of poultry to be treated}} = \text{.... mg product per litre drinking water}$$

Daily water consumption (litre) per animal

* 24mg TMP + sulfa / kg body weight is equivalent to 32mg veterinary drug per kg body weight

Preparation of drinking water medicated

Half fill the container with potable water, add the prescribed amount of product, shake vigorously, then add the rest of the water.

In order to ensure the correctness of the dose administered, body weight should be determined as precisely as possible.

The daily amount should be added to the drinking water so that the required medication is administered within 24 hours. Medicated drinking water and stock solutions should be replaced every 24 hours. Animals should not have access to other sources of water while they are being administered medicated water. The dilution should be calculated so that the animals are sufficiently watered. At the end of the treatment, care should be taken to clean the drinker properly to prevent ingestion of the active substance in sub-therapeutic amounts. Body weight and water consumption should be determined as precisely as possible to ensure administration of the appropriate dose. The consumption of medicated water depends on the clinical condition of the animals.

Route of Administration

For oral administration via drinking water.

Contraindication

Do not use in hypersensitive animals or animals with liver or kidney disease.

Do not administer to animals whose hematopoietic function is affected. Do not use in case of known hypersensitivity to sulfonamides, trimethoprim or any of the excipients.

Warning and Precautions

Special warnings for each target species:

Avoid underdosing. Sick animals can drink less water than normal. If animals can be isolated, the concentration of the veterinary medicinal product can be increased proportionally in water. The watering should be checked regularly, especially in broilers.

Special precautions for use

Special precautions for use in animals:

Due to the variable sensitivity (in time and space) of bacteria to potentiated sulfonamides, the resistance of bacteria is likely to fluctuate from one region to another, or even from one farm to another. Consequently, it is recommended to take bacteriological samples and perform sensitivity tests. The administration of the veterinary medicinal product in question must be carried out according to the culture and the sensitivity of the microorganisms in sick farm animals or according to previous recent experiences on the site concerned. If the veterinary medicinal product is used in other ways than those indicated, we cannot rule out any risk of increasing the prevalence of bacteria resistant to sulfachloropyridazine and trimethoprim and of reducing the effectiveness of combinations of trimethoprim with other sulfonamides due to cross-resistance. During any use of the veterinary medicinal product, the official and local antimicrobial policy must be respected.

Special precautions to be taken by the person administering the veterinary medicinal product to animals:

Sulfonamides can cause hypersensitivity reactions (allergy) after injection, inhalation, ingestion or skin contact. Hypersensitivity to sulfonamides may cause cross reactions with other antibiotics. The allergic reactions caused by these substances can be severe in some cases.

Avoid contact with this veterinary medicinal product if you are sensitive to sulfonamides. If after exposure to the veterinary medicinal product symptoms such as a rash appear, seek medical advice immediately and show the package leaflet to the doctor. Put on a pair of waterproof latex or rubber gloves and put on a pair of safety glasses before using this veterinary medicinal product.

In case of eye contact, rinse with plenty of clean water and, if irritation occurs, call a doctor.

In case of accidental ingestion, consult a doctor.

Wash hands and contaminated skin immediately after handling the veterinary medicinal product.

Interaction with Other Medicaments

Do not combine with other veterinary medicinal products.

Pregnancy and Lactation

The safety of the veterinary medicinal product has not been established during lay.

Side Effects

Sulfonamides can affect kidney function (crystalluria, hematuria, urinary obstruction) and hematopoiesis (thrombocytopenia, anemia).

The frequency of side effects is defined as follows:

- very common (side effects in more than 1 animal in 10 animals treated)
- common (more than 1 but less than 10 animals in 100 animals treated)
- uncommon (more than 1 but less than 10 animals in 1,000 animals treated)
- rare (more than 1 but less than 10 animals in 10,000 animals treated)
- very rare (less than 1 in 10,000 animals treated, including isolated cases)

Symptoms and Treatment of Overdose

The target species support doses at least 5 times higher than the recommended dose which is administered during a treatment period at least three times greater than the maximum recommended period, without any adverse effects having been demonstrated.

Storage Condition

Store below 30°C.

Shelf life

3 years.

To be used within 24 hours after dilution in drinking water.

After first opening the container, any product remaining after 24 hours should be discarded.

Withdrawal Period

Chickens: Meat and offal: 7 days

Eggs: Do not use in egg laying hens intended for human consumption.

Packing

100g, 200g, 500g, 1kg.

Revised date: 2.9.2020

	Manufacturer & Product Registration Holder
	Range Pharma Sdn Bhd No. 1, Jalan TSB 11, Taman Industri Sg. Buloh, 47000 Sg. Buloh, Selangor Darul Ehsan, Malaysia Tel: 603-61568708 Fax: 603-61568707 Email: info@rangepharma.com